Non-Verbal Reasoning

12. HOLE PUNCH / TYPE 12

In these questions you will be given a shape is folded several times, shown in a sequence. After the final fold, a hole is punched through the shape. You have to work out what the shape will look like when it is unfolded again. Below is a worked example. The explanation shows how the hole punched shape will look at each stage of the unfolding.

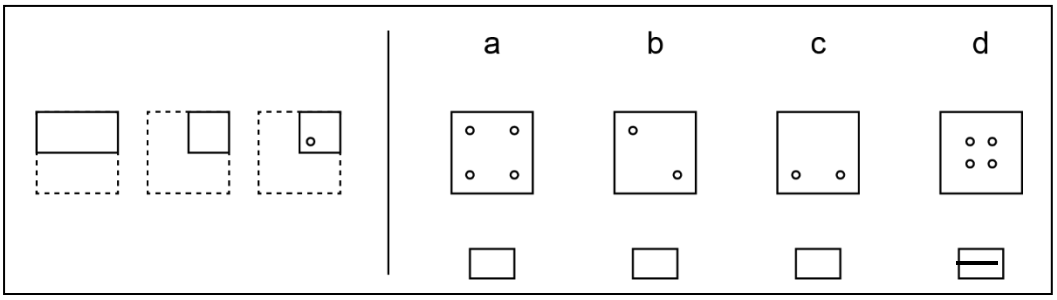
What to look out for

- This question type is a test of logic and spatial awareness. You need to take a step by step approach to working out the answer.
- Carefully study how the shape has been initially folded. Imagine it as a piece of paper. It is vital that you understand this part of the question first before you start looking at the hole punch.
- Pay careful attention to where the hole punch occurs and its position within the folded shape.

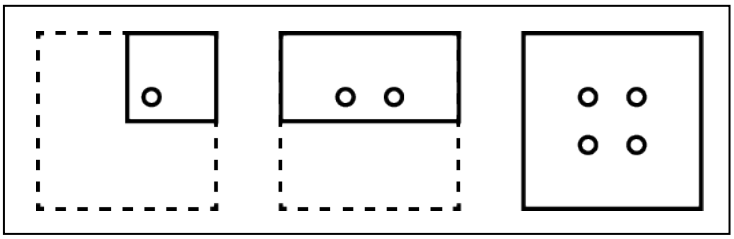
Technique tips

- The initial hole punch is a really useful clue! In the example below, you know that one hole <u>must</u> be just off-centre in the top right corner because the hole punch will go all the way through to the bottom layer. You can therefore work out straight away that B and C cannot possibly the correct answer.
- Using the sequence given in the question, think about how the shape will look after the first unfold. Will the hole punch have 'gone through' to the unfolded side?
- In the example below, we know that the original square has been folded over itself completely twice. This means that the shape is now 4 layers thick (folding up once has created 2 layers, and folding across once has doubled the layers to 4). We then know that 1 hole through 4 layers must result in 4 holes when unfolded.
- However, be careful with this approach. In some questions, the shape is only partially folded over itself. You must then think carefully about the position of the hole punch and whether it will 'go through' all layers.

Example



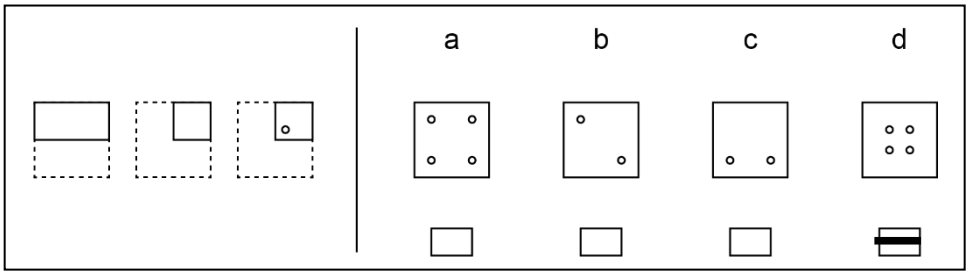
Explanation



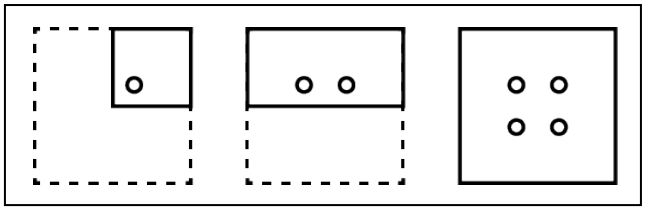
Non-Verbal Reasoning

12. HOLE PUNCH / TYPE 12

Example



Explanation



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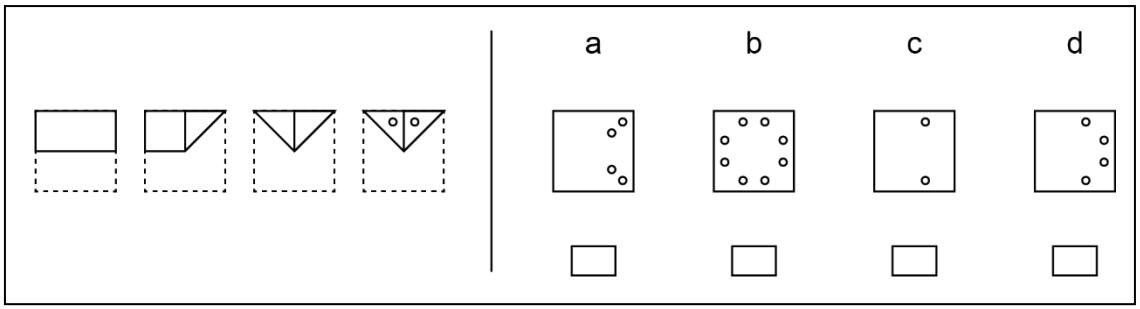

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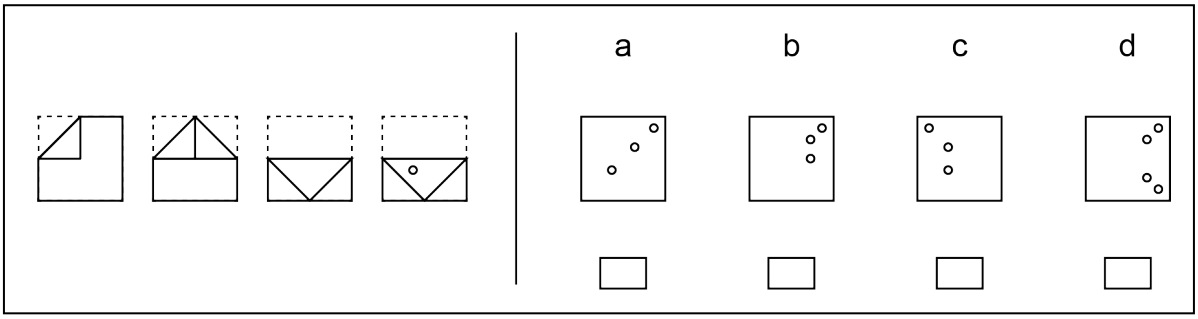

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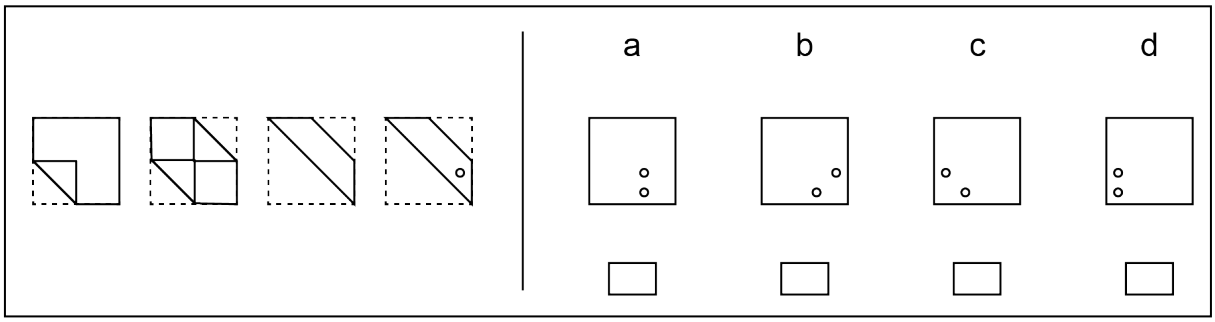

Non-Verbal Reasoning

12. HOLE PUNCH / TYPE 12

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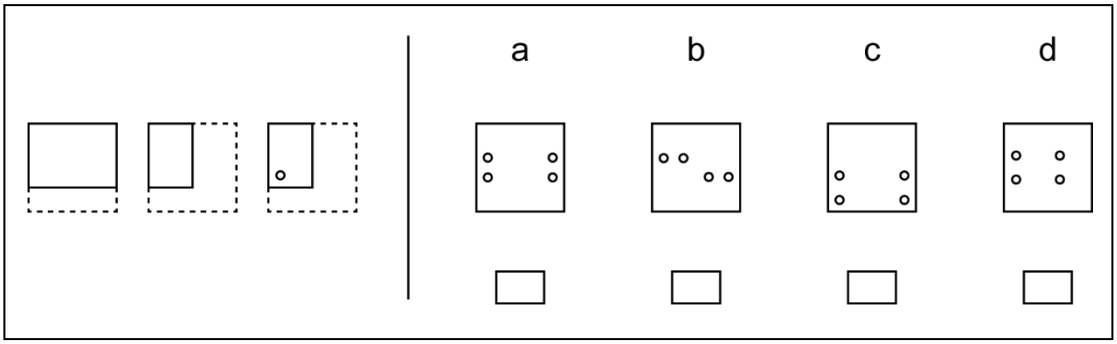

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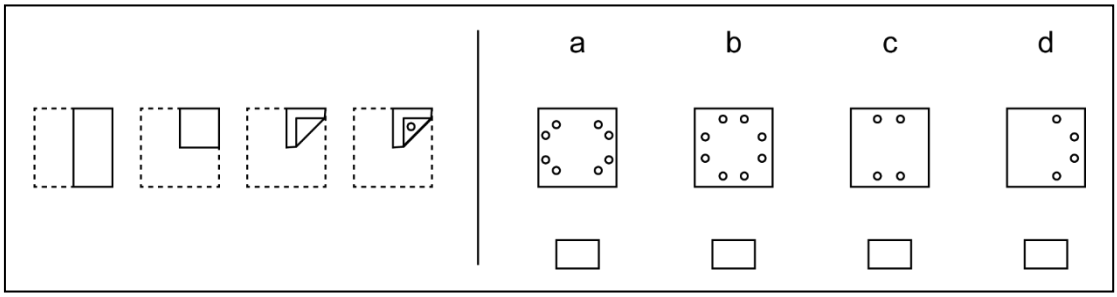

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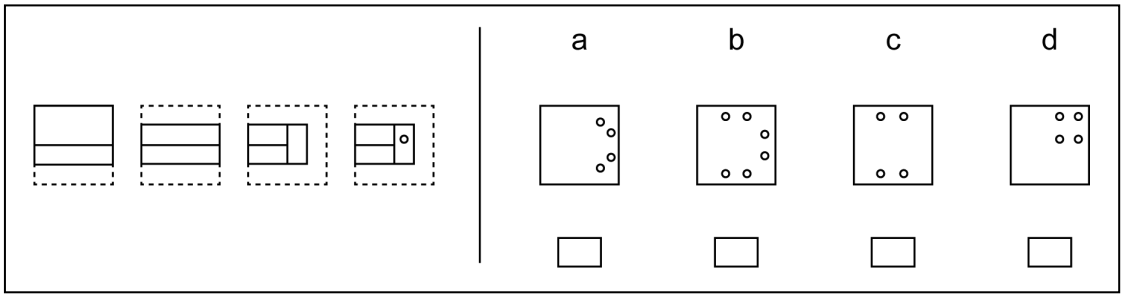

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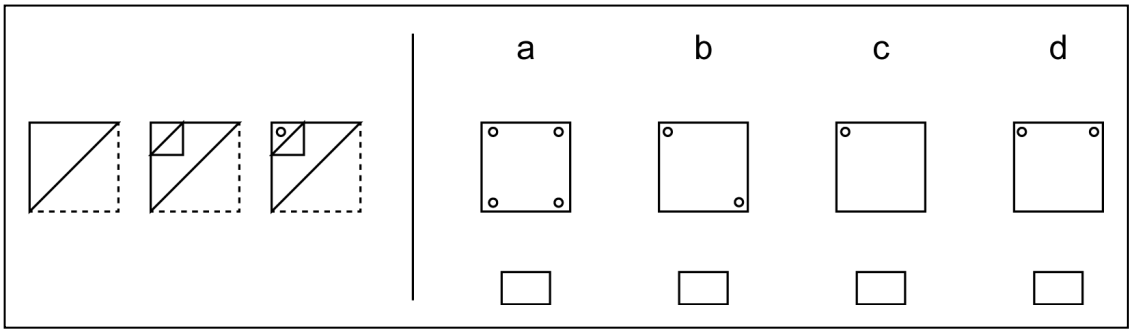

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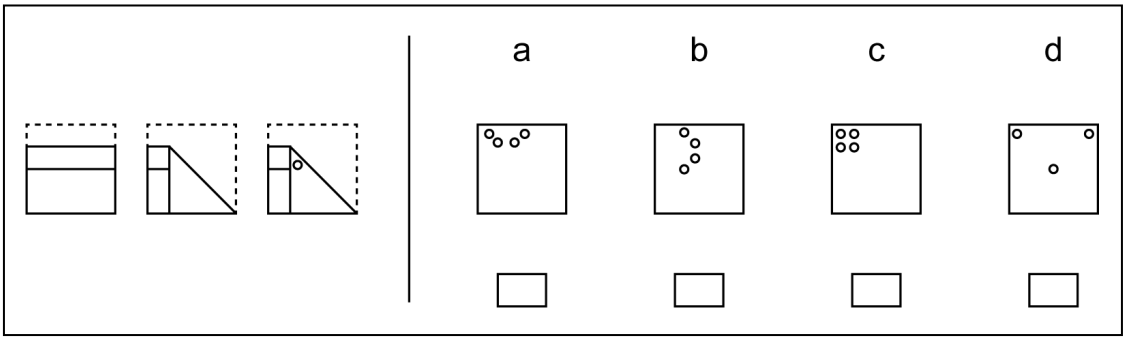


Non-Verbal Reasoning

13. SHADED NETS / TYPE 13

In these questions you will be given the net of a 3D shape that has been shaded in some way. You have to work out which of the answers correctly represents the 3D shape that will be created from the net given. Below is a worked example.

What to look out for

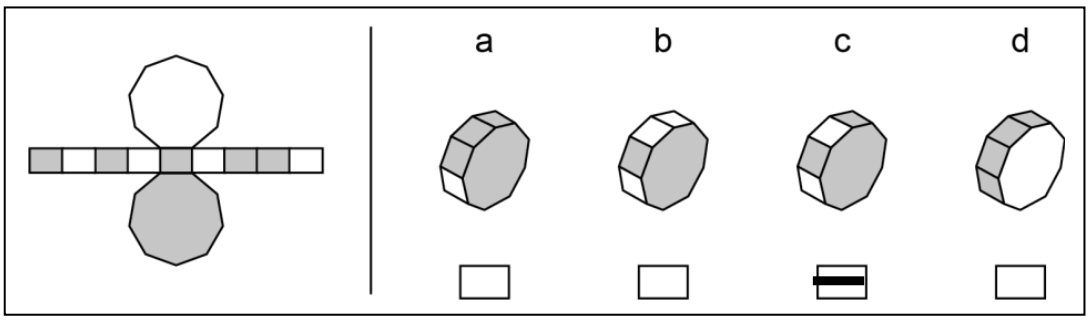
As with all Non-Verbal Reasoning question types this is largely a test of **logic** and **close observation**.

- Take a careful note of how the net is shaded. Is there any pattern?
- Remember that a net can sometimes be folded in more than one way to create a shape. In the example below, the two larger faces could be folded up or down. In different questions this may give different possible answers.

Technique tips

- Have an open mind with these questions. You may have to try to visualise what the shape created will look like from different viewpoints.
- Use the net to work out how the different faces should look in relation to one another. In the example below, you know that there are never two white rectangular faces next to each other.
- If you are struggling to see the correct answer immediately, use logical deductions to eliminate answers that are definitely incorrect. This can be seen in the explanation below.
- In some questions, the faces of the shape will be different sizes. Take note of whether the larger faces are shaded or white, as they may be easier to spot in the answer options.

Example



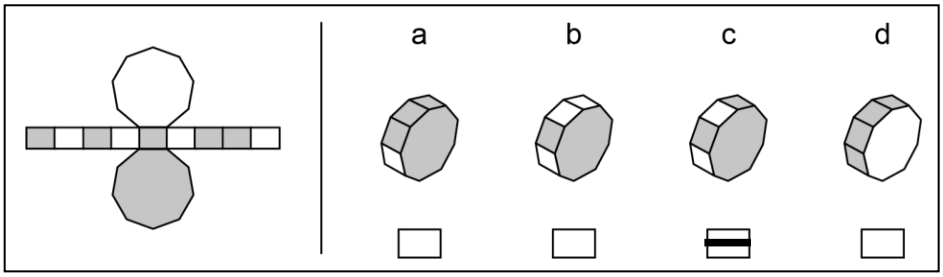
Explanation

The answer cannot be A because in the net three shaded rectangles do not appear next to each other consecutively. The answer cannot be B because in the net two white rectangles do not appear next to each other. The answer cannot be D because in the net four shaded rectangles do not appear next to each other consecutively. Therefore the answer is C.

Non-Verbal Reasoning

13. SHADED NETS / TYPE 13

Example



Explanation

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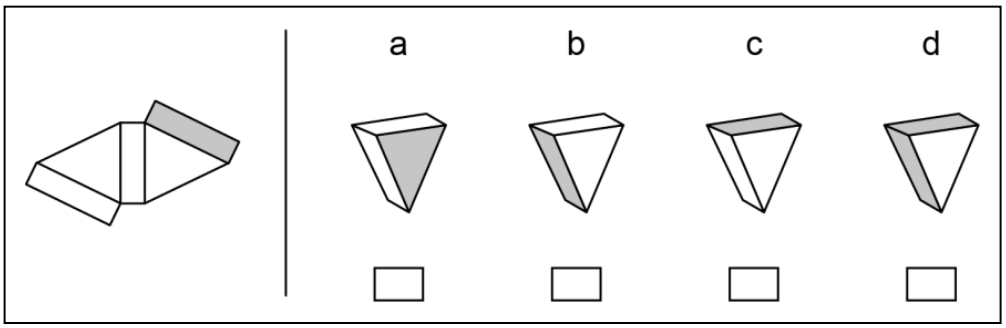

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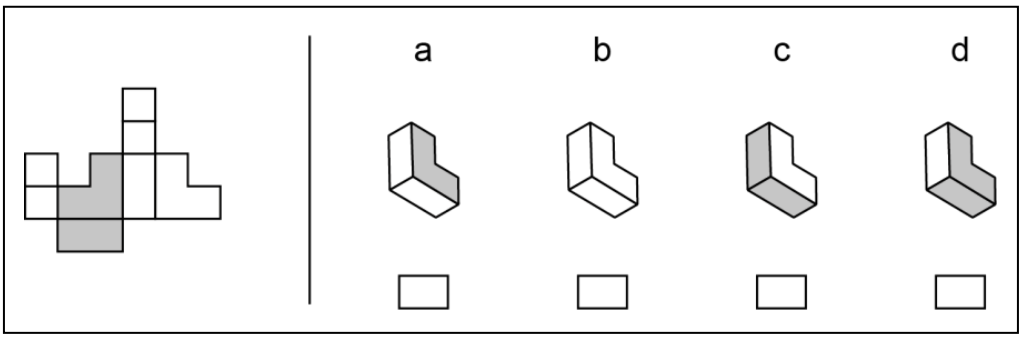

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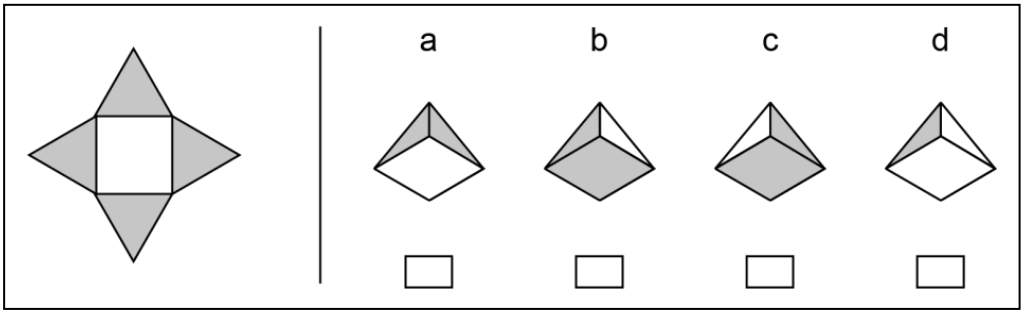
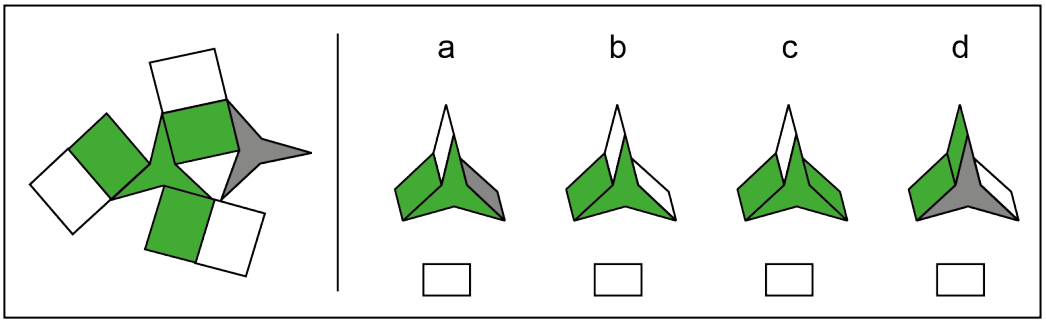


Non-Verbal Reasoning

13. SHADED NETS / TYPE 13

4



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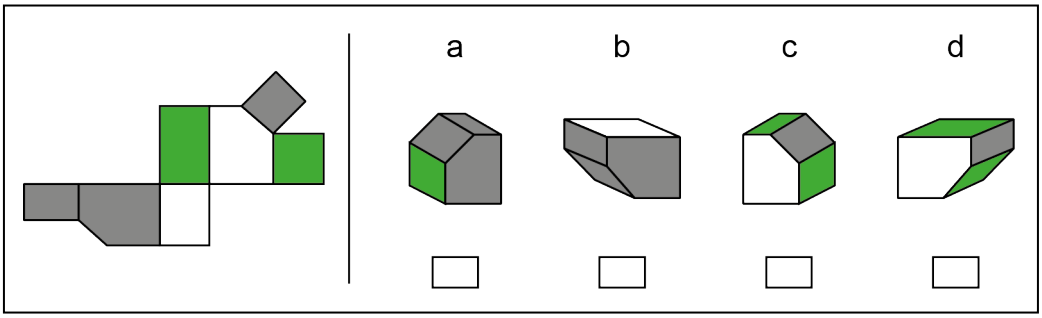

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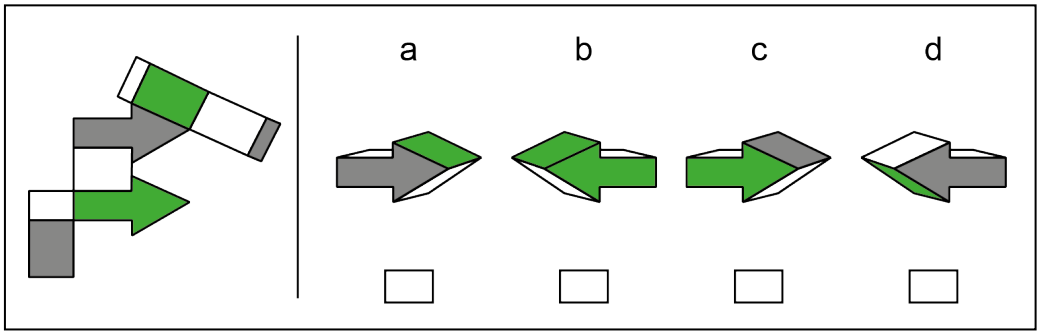

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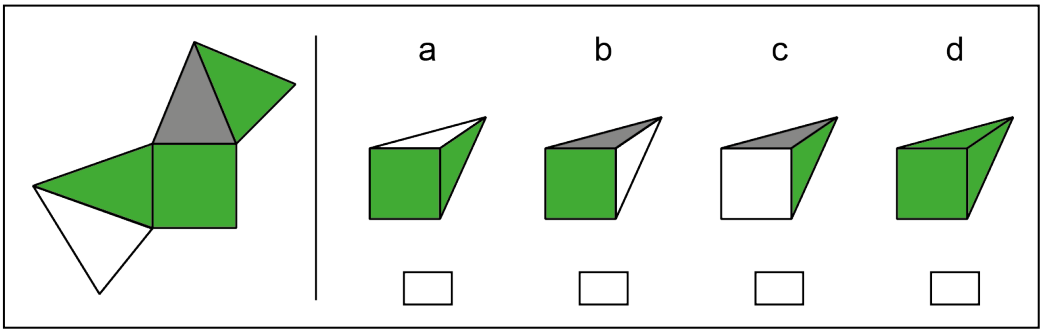

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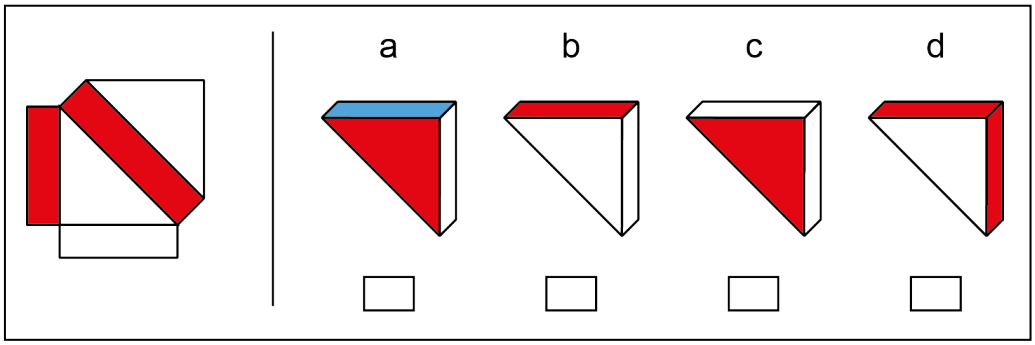
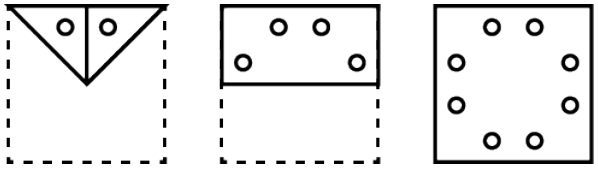


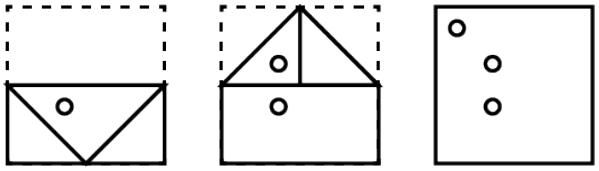
Non-Verbal Reasoning

Hole Punch

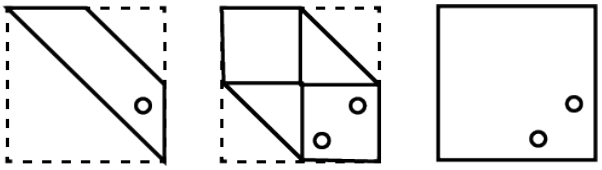
1. **B** -



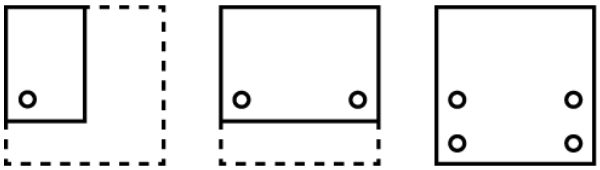
2. **C** -



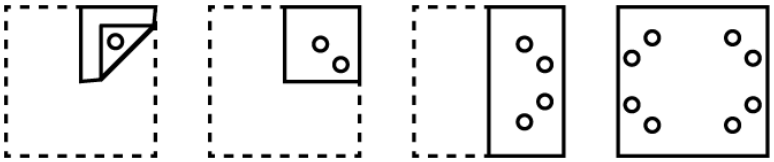
3. **B** -



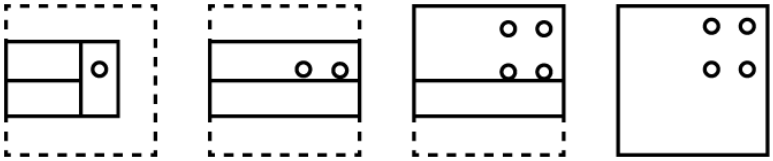
4. **C** -



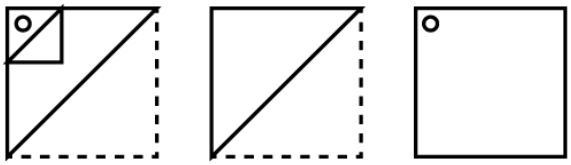
5. **A** -



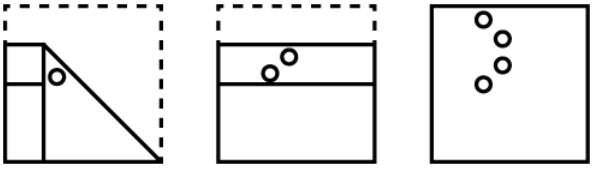
6. **D** -



7. **C** -



8. **B** -



Shaded Nets

1. **B** - (B is the correct answer because it is the only option that shows just one of the longer rectangles shaded and the rest of the sides white.)
2. **D** - (The correct answer must have one of the 'L' shaped faces and one of the longer rectangular faces shaded. D is the only option that satisfies these criteria.)
3. **A** - (The correct answer must have a white square base and all the visible triangular faces must be shaded.)
4. **B** - (From the net you can tell that the rectangular faces alternate in colour between green and white. Once you spot this you can quickly tell that A and D definitely cannot be correct. You can then work out that B must be correct rather than C.)
5. **A** - (The answer cannot be B because the largest rectangular face is the wrong colour. The answer cannot be C or D because no matter which way the net is folded the rectangular faces cannot alternate in colour green, grey, green. Therefore A is the correct answer.)
6. **D** – (The answer cannot be A because if the grey arrow face were pointing right the top flat rectangular face would be grey. The answer cannot be B because B would be the reverse of A and therefore the top flat rectangular face would again have to be grey. The answer cannot be C because if the green arrow face were pointing right the top angled rectangular face would have to be white. Therefore the answer is D.)
7. **A** – (The square base must be green, therefore the answer cannot be C. The pattern of triangular faces is white, green, grey, green. The only answer that matches this pattern is A.)
8. **B** – (The triangular faces must both be white, therefore the answer must be either B or D. You can see from the net that one of the shorter rectangular faces must be red while the other shorter rectangular face must be white. Therefore the correct answer must be B rather than D.)